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Railway applications — Compatibility between rolling stock and train detection systems

The European Standard EN 50238:2003 has the status of a
British Standard

ICS 29.180; 45.060.10

National foreword

This British Standard is the official English language version of EN 50238:2003.

The UK participation in its preparation was entrusted by Technical Committee GEL/9, Railway electrotechnical applications, to Subcommittee GEL/9/1, Signalling and communications, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
- monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

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EUROPEAN STANDARD

EN 50238

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EUROPÄISCHE NORM

February 2003

ICS 29.180; 45.060.10

English version

**Railway applications –
Compatibility between rolling stock
and train detection systems**

Applications ferroviaires –
Compatibilité entre matériel roulant
et systèmes de détection de train

Bahnanwendungen –
Kompatibilität zwischen Fahrzeugen
und Gleisfreimeldesystemen

This European Standard was approved by CENELEC on 2002-12-01. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CENELEC member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CENELEC member into its own language and notified to the Central Secretariat has the same status as the official versions.

CENELEC members are the national electrotechnical committees of Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Slovakia, Spain, Sweden, Switzerland and United Kingdom.

CENELEC

European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

Central Secretariat: rue de Stassart 35, B - 1050 Brussels

Foreword

This European Standard was prepared by SC 9XA, Communication, signalling and processing systems, of Technical Committee CENELEC TC 9X, Electrical and electronic applications for railways.

The text of the draft was submitted to the formal vote and was approved by CENELEC as EN 50238 on 2002-12-01.

This European Standard was prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association and supports the essential requirements of Directive 96/48/EC.

The following dates were fixed:

- latest date by which the EN has to be implemented
at national level by publication of an identical
national standard or by endorsement (dop) 2003-12-01
- latest date by which the national standards conflicting
with the EN have to be withdrawn (dow) 2005-12-01

This European Standard is intended to be read in conjunction with the EN 50121 series.

Annexes designated “informative” are given for information only.
In this standard annexes A, B, C and D are informative.

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Introduction

This European Standard defines a process to obtain the assurance that specific rolling stock operating on a specific route does not interfere with train detection systems installed on this route.

Compatibility problems between train detection systems and rolling stock are a significant obstacle to cross-acceptance of rolling stock in Europe. Unfortunately it is not possible to establish general rules for the maximum levels of interference allowed, valid for every country. This is due to the great diversity of rolling stock, power supply and return current systems, and train detection systems installed in Europe. This diversity leads to consideration of the problem of compatibility of rolling stock and train detection systems for specific routes to avoid unnecessarily restrictive specifications.

Compatibility is determined by both physical and electromagnetic considerations. With regard to EMC, the need is not for general values for maximum levels of interference permitted, but for convenient methods by which to specify the level of interference allowed for operation on specific routes.

Interference may be caused by

- rail currents,
- electromagnetic fields,
- differential voltage between axles,

as shown in Figure 1:

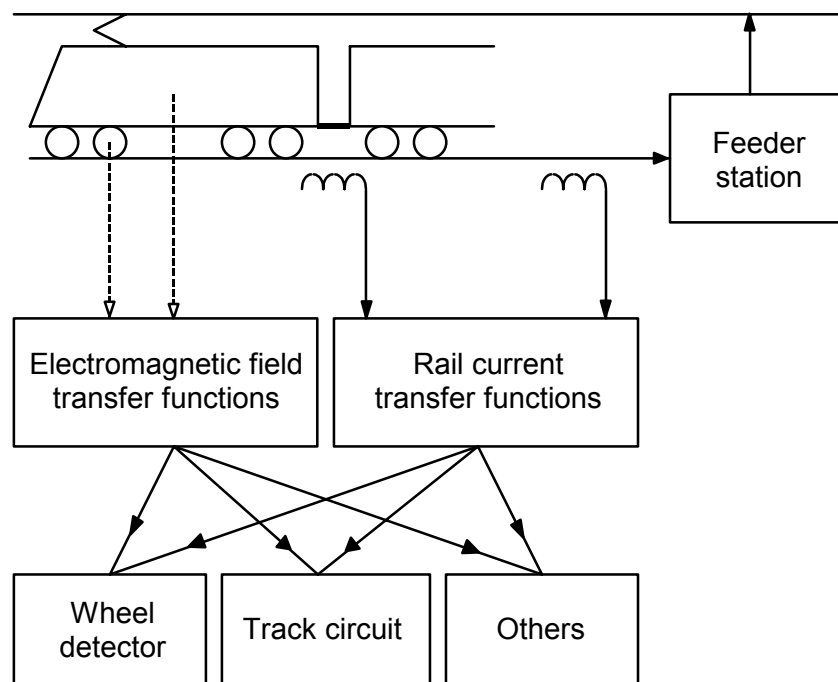


Figure 5 – Sources of electromagnetic interference

In practice, the susceptibility of the system is determined by

- the sensitivity of individual components of the system,
- the application of the components, i.e. the configuration of the system.

Therefore the problems concerning track circuits and axle counters or wheel detection systems will be looked at separately.

For determining the susceptibility of signalling systems, laboratory/simulation testing methods as well as methods to conduct tests on the “real railway” are proposed. Modelling enables worst-case conditions to be simulated. In addition, particular test sites are used because, from experience, they are known to provide the test evidence required. Then, taking account of the experience of the railways, it is possible to establish a general method for determining the susceptibility of train detection systems, described in this European Standard.

Before measuring the interference level on rolling stock, a sufficient knowledge of the electric circuit diagram of the power equipment is required, e.g. switching frequencies of on-board static converters, type of regulation used for power converters, resonant frequency of each filter, operating limits under high and low supply voltages, downgraded modes of operation etc.

1 Scope

The scope of this European Standard is to describe a procedure for mutual acceptance of rolling stock to run over specific routes. It describes the methods of measurement of interference currents, the methods of measurement of the susceptibility of train detection systems, the characterisation of traction power supplies and the procedure for acceptance. The result of the acceptance procedure is a structured justification document referred to as a “compatibility case”, which documents the evidence that the conditions for compatibility have been satisfied.

This European Standard is not generally applicable to those combinations of rolling stock, traction power supply and train detection system which were accepted as compatible prior to the issue of this European Standard. However, as far as is reasonably practicable, this European Standard may be applied to modifications of rolling stock, traction power supply or train detection systems which may affect compatibility.

The scope of the compatibility case is restricted to the demonstration of compatibility of rolling stock with a train detection system’s characterisation (e.g. gabarit). Radio based signalling systems are not within the scope of this European Standard.

2 Normative references

This European Standard incorporates by dated or undated references, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed below. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

EN 50121	Series	Railway applications – Electromagnetic compatibility
EN 50126		Railway applications – The specification and demonstration of Reliability, Availability, Maintainability and Safety (RAMS)
EN 50163		Railway applications – Supply voltages of traction systems
EN/ISO 9001		Quality management systems – Requirements
EN ISO/IEC 17025		General requirements for the competence of testing and calibration laboratories
ORE B108/1		Unification of air-conditioning and electrical equipment in coaches
UIC 737-3		The application of thyristors in railway technology: Measures for the prevention of functional disturbances in signalling installations
UIC 550		Power supply installations for passenger stock

3 Definitions

For the purposes of this European Standard, the following definitions apply:

3.1

accepting body

the body responsible for the evaluation of the compatibility case and the issue of a certificate of acceptance

3.2**compatibility case**

a suite of documents which records the evidence demonstrating the degree of compatibility between rolling stock, traction power supplies and train detection systems for a specific route or specific railway network

3.3**certificate of acceptance**

written authorisation from the accepting body that the compatibility case is acceptable to allow the new or modified systems to enter service (This may be qualified)

3.4**degraded modes**

modes of operation in the presence of faults which have been anticipated in the design of the rolling stock. Degraded modes will normally allow the rolling stock to complete its journey

3.5**gabarit**

the maximum permissible levels of interference signal, with respect to frequencies and duration, to which a train detection system may be exposed

3.6**railway infrastructure authority**

the body responsible for the safety of the track and signalling systems

3.7**right side failure**

a failure of a signalling system which results in a more restrictive condition for the movement of traffic than is appropriate

3.8**rolling stock operator**

the body responsible for the operation and maintenance of the rolling stock

3.9**wheel detector**

a sensor which detects the passage of a wheel. It may be used as part of an axle counter system or as a treadle

3.10**wrong side failure**

a failure of a signalling system which results in a less restrictive condition for the movement of traffic than is appropriate

4 Acceptance process**4.1 Overview**

The parties concerned in the acceptance process are shown in Figure 2:

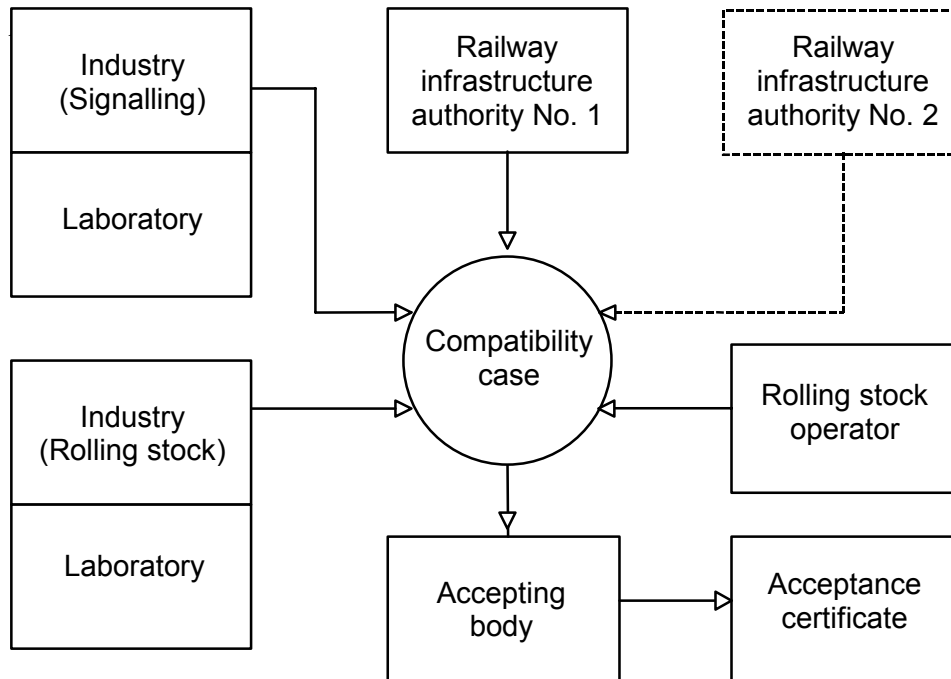


Figure 6 – The parties concerned in the acceptance process

4.2 Responsibilities

The responsibility for demonstrating compatibility between rolling stock, train detection and traction power supply systems, and for maintaining it over the full life cycle of the equipment, is shared between those parties responsible for the particular railway infrastructure and the particular rolling stock. Specific responsibilities for a given compatibility case, including the party taking the lead role, shall be explicit. The documentation, in the form of a compatibility case, shall be submitted to an accepting body, and shall be reviewed when any modification is carried out.

4.2.1 Railway infrastructure authority

For a defined route (the application of interest), the railway infrastructure authority should characterise all train detection systems and the traction power supply system.

4.2.2 Rolling stock operator

The rolling stock operator should characterise the interference which may be generated and propagated by the rolling stock.

4.2.3 Accepting body

The accepting body shall review submitted documents and as a result issue a certificate of acceptance. As part of this process, the accepting body should ensure that the compatibility case is reviewed by experts who are qualified to assess and evaluate it.

4.3 Acceptance process

The acceptance process is summarised in Figure 3:

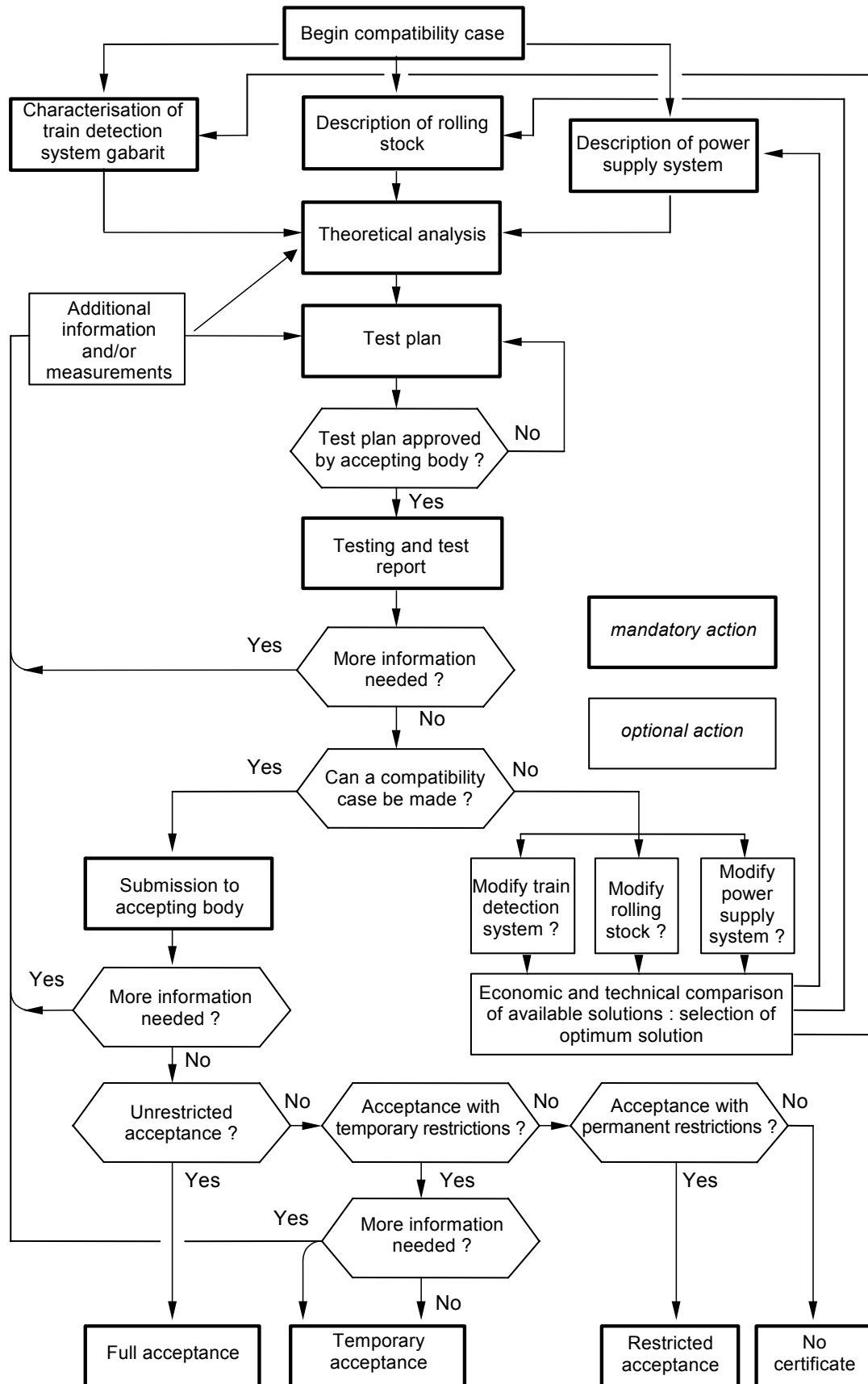


Figure 7 – The acceptance process

4.4 Compatibility case

A compatibility case shall be prepared, including but not limited to, the following:

- scope of compatibility case;
- route identification;
- characterisation of train detection systems;
- characterisation of traction power supply system;
- characterisation of rolling stock;
- assumptions made;
- test report;
- evidence of quality management;
- related compatibility cases;
- assessment of compatibility.

The compatibility case shall be submitted to an accepting body for approval.

4.5 Quality management

Quality management systems shall be in place. The importance of configuration management should be noted.

The configuration state of the relevant infrastructure and rolling stock (including maintenance processes and schedules) shall be recorded and referenced within the compatibility case. Any subsequent changes to these configurations shall lead to an examination of the continued validity of the compatibility case.

The organisation conducting the tests shall be able to demonstrate its competence to make measurements in a traction environment. It should preferably be certified to EN ISO/IEC 17025, but in any case shall have a documented quality system in accordance with a recognised standard.

In order to maintain objectivity (e.g. in the event that tests are being conducted by a manufacturer on equipment of his own supply) the organisation conducting tests should be subject to an audit by the accepting body.

4.6 Route identification

In order to accept a particular rolling stock in respect of a particular route or network, the different types and applications of train detection systems and traction power supply systems on the route and on adjacent routes which may be affected shall be identified. In addition to the intended operational route(s), alternative route(s) which may be required in the event of disruption to traffic shall also be considered.

4.7 Characterisation

The characteristics of the identified systems shall be obtained in accordance with the following clauses:

- train detection systems 5;
- rolling stock 6;
- power supply system 7.

4.8 Tests

A test plan shall be prepared in accordance with 6.4.

Tests shall be conducted in accordance with the test plan and a test report produced in accordance with Clause 6.

4.9 Compatibility analysis

In general terms, it shall be demonstrated that the rolling stock characteristics for generated and propagated interference comply with the train detection system gabarit, under defined operating conditions, including degraded modes. Their relationship is shown in Figure 4. The information flow may be in either direction depending on which system is to be changed.

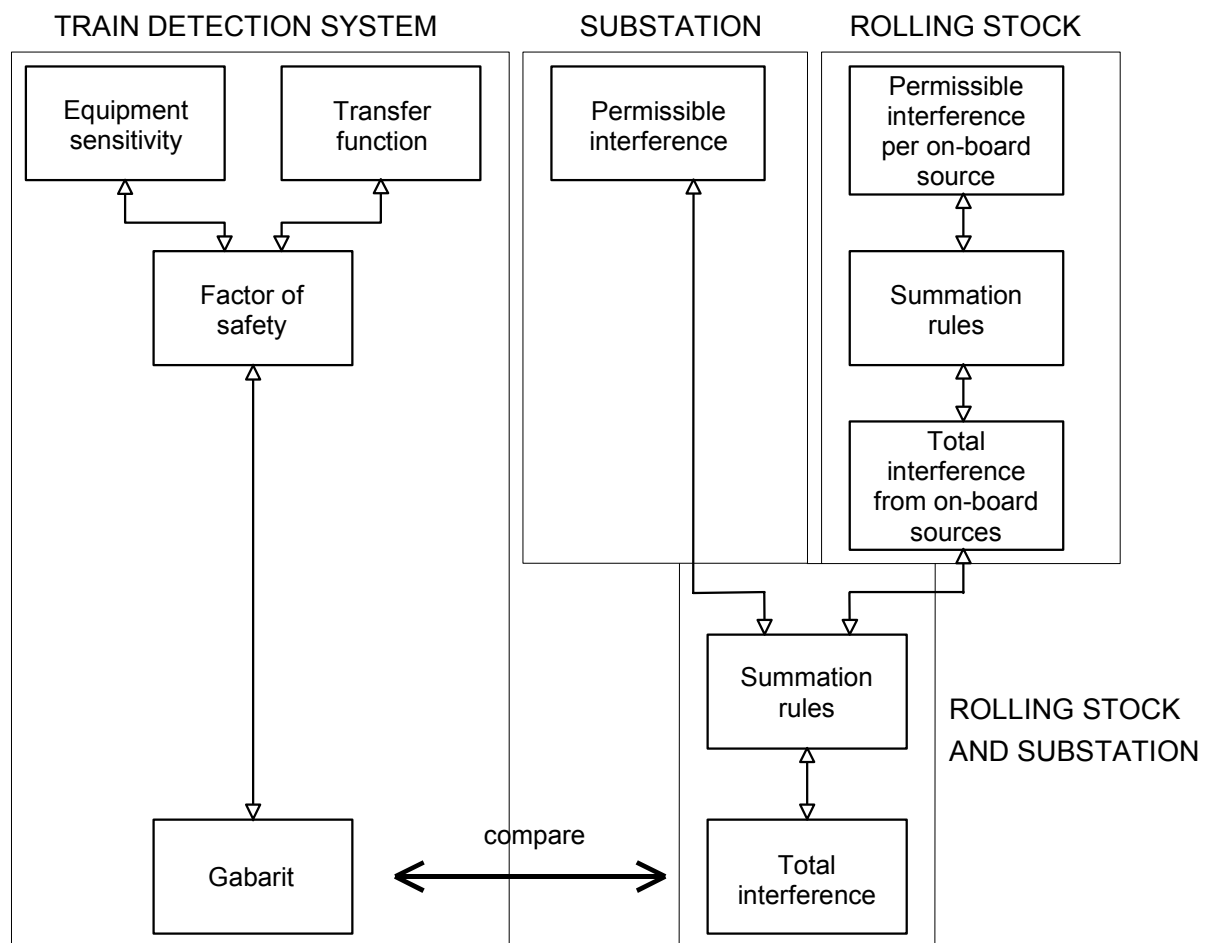


Figure 8 – Relationship between gabarit and permissible interference

Additionally, the physical characteristics of the rolling stock shall be demonstrated to be compatible with the train detection systems.

The compatibility analysis is mandatory and shall explain the technical principles which assure compatibility, including (or giving reference to) all supporting evidence e.g., calculations, test plans and results etc.

The method of analysis of fault modes shall be agreed between the parties listed in 4.2. If the parties consider it to be necessary, for complex systems, reference may be made to the clauses in EN 50126 entitled “risk” and “risk analysis”.

NOTE In accordance with the new organisation defined by EC Directive 91/440, it could be necessary to define a new "infrastructure gabarit" including both the train detection system gabarit and the substation characteristics.

4.10 Certificate of acceptance

The accepting body shall review the compatibility case and issue a certificate of acceptance.

It is recognised that characterisation of interference generated and propagated by rolling stock can be a time consuming process, which may require a significant amount of testing during service operations in order to refine the characteristics. Therefore, provided that the risks to all parties can be demonstrated to be acceptable, temporary acceptance may be given prior to full acceptance. This also allows for the identification of contributory factors which had been overlooked in the preparation of the compatibility case.

Any temporary acceptance shall be time limited and be provided for a specific agreed purpose, whilst additional measures to identify interference and mitigate against possible hazards should be implemented.

In the event that the assessment identifies either a non-compliance with the existing gabarit, or a worsening of the existing gabarit (e.g. higher susceptibility), such that rolling stock with existing route acceptance becomes non-compliant, then the relevant Railway Infrastructure Authorities and rolling stock operators shall jointly determine the modifications to infrastructure and/or rolling stock to be made and initiate acceptance processes where applicable. If modifications are not possible or are not carried out, permanent restrictions shall be applied.

5 Characterisation of train detection systems

5.1 Objective of procedure

To ensure the correct operation of train detection systems, their physical and electromagnetic properties shall be checked against those of the rolling stock and the traction power supply system.

5.2 Physical compatibility

5.2.1 Track circuits

Physical compatibility, including but not limited to the following aspects, shall be considered:

- minimum track circuit length;
- minimum train length;
- maximum speed of trains;
- response time of track circuits;
- shunt values of track circuits;
- shunt impedance of trains, and reliability of shunting in all service conditions;
- axle loads of vehicles;
- axle spacing;
- body overhang;
- track characteristics;
- substations earthing arrangements.

5.2.2 Wheel detectors

The reliable detection of wheels may be influenced by factors including but not limited to the following:

- wheel geometry;
- speed of trains;
- wheel material;
- equipment which may be mistaken for a wheel, e.g. track brakes or metal assemblies which are mounted close to the rail head.

5.2.2.1 Wheel dimensions and material

Axle counters and treadles are generally specified to detect reliably wheel types used on main line vehicles. Other vehicle types may require special settings of the wheel detectors.

5.2.2.2 Interfering equipment

Equipment mounted within the immediate vicinity of the rail may interfere with the reliability of wheel detection either by fouling mechanically or due to their electromagnetic properties (see Annex A).

This aspect of compatibility shall be assessed by testing.

5.3 Electromagnetic compatibility

This subclause describes the measurement of the gabarit of the train detection system. It defines the general approach to be adopted, but since every installation is different in detail, it cannot describe the whole process exactly. The compatibility case shall cover all credible configurations and parameters – some examples of configuration and models are given in Annex A.

The required measurements are as follows.

5.3.1 The sensitivity of the train detection system equipment (see 5.3.3.1 for track circuits; 5.3.4.1 for wheel detectors).

5.3.2 The “transfer function” relating the interference signal at the train detection system equipment to the interference generated by the rolling stock (see 5.3.3.2 for track circuits; 5.3.4.2 for wheel detectors).

Let the transfer function be denoted by F .

Let the interference signal at the train detection system equipment be denoted by I_{TDS} .

Let the interference signal generated by the rolling stock be denoted by I_{RS} .

The interference signal is then:

$$I_{TDS} = F \cdot I_{RS}$$

Hence

$$I_{RS} = I_{TDS} / F$$

The maximum permissible interference signal at the train detection system equipment I_{TDSmax} is determined by the sensitivity of the train detection system equipment. Let the total permissible interference generated by rolling stock be denoted by I_{RStot} . Then:

$$I_{RStot} = I_{TDSmax} / F$$

Where multiple sources may contribute to the total interference signal, the permissible interference per source shall take this into account (see 5.8).

Note that the permissible interference signal will have two values determined by the following criteria:

- the signal which may cause the train detection system to show clear when it is in fact occupied (a wrong side failure, i.e. a matter of safety);
- the signal which may cause the train detection system to show occupied when it is in fact clear (a right side failure, i.e. a matter of reliability). The effect on interlocking logic shall however be considered.

5.3.3 Track circuits

5.3.3.1 Sensitivity and susceptibility of equipment alone

The sensitivity of the track circuit equipment itself shall be determined. In general the susceptibility of the receiver will be more significant than that of the transmitter and will be the determining factor in the permissible level of interference; however the susceptibility of the transmitter shall also be checked.

Where the sensitivity of the track circuit is adjustable, it shall be measured for all the relevant settings and in particular for the worst case settings.

The susceptibility shall be determined as follows.

5.3.3.1.1 The parameters (amplitudes, frequencies and durations) of the voltage(s), current(s) or electro-magnetic fields which may cause the receiver to be energised, in the absence of a signal from the transmitter, such that a “track circuit unoccupied” indication is given.

NOTE Some types of track circuit receiver which are designed to operate with an amplitude modulated signal may also be energised by a combination of unmodulated frequencies, and some types of track circuit receiver which are designed to operate with a frequency modulated signal may also be energised by one or more amplitude modulated frequencies.

5.3.3.1.2 The parameters (amplitudes, frequencies and durations) of the additional voltages, currents or electro-magnetic fields which may de-energise the receiver, or may influence the output signal of the transmitter, such that a “track circuit occupied” indication is given.

The above information shall be obtained from the suppliers of the track circuit equipment. If it is not available (for example in the case of an obsolete design) it shall be obtained by laboratory measurements.

5.3.3.2 Transfer function of track circuit as installed

The transfer function of the installed track circuit with regard to interference current shall be determined as follows.

5.3.3.2.1 Determine the electrical equivalent circuit of the track circuit and the traction power supply system in so far as the latter may affect the track circuit. In doing this all relevant conductors shall be considered, including catenaries or conductor rails, cross-bonded running rails, return conductors, impedance bonds, booster transformers, earthed structures, earth paths, etc. Normally the “worst case” transfer function will correspond to the maximum length track circuit.

5.3.3.2.2 Set up the equivalent circuit using a real test site or a hardware or software model. If a model is used it shall be verified by means of comparative site tests.

5.3.3.2.3 Measure the voltage or current at the track circuit equipment terminals resulting from a current generated by the rolling stock. The transfer function is then the ratio of the voltage at the track circuit equipment terminals to the interference current.

NOTE 1 The value of the transfer function will depend on the position of the train with respect to the track circuit. The worst case value shall be determined.

NOTE 2 If the train has two or more on-board sources, or is composed of electric multiple unit stock, or utilises the rails for the return current path for auxiliary power supplies (see UIC 550), it may have to be represented by more than one current source.

5.3.3.2.4 Repeat the above tests with fault conditions applied, for example broken rails, broken cross-bonds and broken return conductors, examples of which are shown in Annex A (see 4.9). The “worst case” transfer function is the ratio of the highest value of interference voltage at the track circuit equipment terminals observed under fault conditions to the interference current(s).

The transfer function of the installed track circuit with regard to electromagnetic fields shall be determined in a similar manner to that for interference currents, except that practical site tests will probably be more appropriate than computer modelling for the generation of useful results. An example is given in Annex A.

5.3.4 Wheel detectors

The wheel detector circuits used in axle counting systems are usually galvanically decoupled from currents flowing in the rails.

Axle counters may be affected by electromagnetic interference coupled via the wheel detectors and the sensitivity of the axle counter is thus largely determined by the susceptibility to electromagnetic interference of the wheel detector used. Depending on the configuration, the axle counter may also be affected by interference on the transmission path between the trackside and an evaluation unit in an interlocking room.

The return current (d.c. or a.c.) in the rail can interfere with permanent-magnetic or inductively operating wheel detectors

- directly, through its effect on the sensors,
- indirectly through its effect on the permeability of the rail.

Direct coupled interference is expected to be most severe at the operating frequency of inductive sensors. Harmonics of rail current shall be considered.

Electromagnetic interference fields from equipment on the rolling stock can also affect the wheel detectors. Major sources of interference include:

- transformers and converters, especially when fitted under the vehicle close to the rails. Harmonic content of interference shall be considered;
- magnetic track brakes and eddy-current track brakes, which are mounted on the vehicle directly over the rail. The interference may be dependent on the train speed and the operational position of the brake;
- transmitter antennas mounted on the vehicle, especially when mounted in the proximity of the rails;
- catenary switching operations and the rupture or corrosion of contact wires causing transient currents in the rails which can interfere with the wheel detectors or the transmission path.

In addition to axle counter applications, wheel detectors are also used for switching functions, e.g. for switching on and off level-crossing protection systems. In these applications the effect of transient interference may be different to that of axle counters and shall be considered separately, but the characterisation of the wheel detector itself is common to both applications.

5.3.4.1 Sensitivity of equipment alone

The sensitivity of the wheel detector equipment (wheel detector including signal evaluation electronic device) shall be determined as follows.

5.3.4.1.1 The parameters (including area of sensitivity, amplitudes, frequencies and durations) of the electro-magnetic fields and rail currents which will maintain the wheel detector in such a condition that passing wheels are not detected.

5.3.4.1.2 The parameters (including area of sensitivity, amplitudes, frequencies and durations) of the electro-magnetic fields and rail currents, which will cause the “wheel present” state of the wheel detector when no wheel is present.

5.3.4.2 Susceptibility of wheel detectors as installed

The permissible value of interference signal, both transient and continuous, caused by currents in the rail or by other sources of interference will depend on the value of the internal signal exceeding the switching point of the electronic evaluation device of the wheel detector.

The transfer function of an installed wheel detector with regard to interference current and interference electromagnetic field depends on the physical principle and the geometric arrangement of the wheel detector and the signal evaluation device. Many different parameters (frequency and amplitude of interfering current; frequency, amplitude and direction of interfering electro-magnetic field) determine the transfer function. The inherent complexity and non-linearity of the transfer function make it unsuitable for use in this context.

Due to the inherent difficulties of designing a sufficiently complete electrical equivalent circuit for a type of wheel detector, field testing, or alternatively laboratory tests injecting interference currents into the rail and applying external electromagnetic fields shall be used for measuring the susceptibility of wheel detectors.

5.4 Factor of safety

To allow for uncertainties in the accuracy of measurements and simulations, the susceptibility of the train detection system as determined above shall be increased by a factor of safety. The uncertainties shall be estimated and the factor of safety shall be sufficient to allow for them (see A.10).

Interference due to DC substation ripple may need to be taken into account in the factor of safety.

5.5 Track circuit susceptibility

5.5.1 Permissible interference signal for right side failure

The permissible value of interference signal in the case of right side failure of the track circuit is determined without faults in the traction return system, since track circuits are not expected to work reliably in the presence of such faults.

Let the transfer function in this case be denoted by F_{norm} .

Let the interference signal which causes the track circuit to show occupied while clear be denoted by I_{TDSocc} .

Then the total permissible value of interference $I_{\text{RStot, occ}}$ for this case is given by:

$$I_{\text{RStot, occ}} = I_{\text{TDSocc}} / F_{\text{norm}}$$

for all combinations of amplitude, frequency and duration of the interference signal.

5.5.2 Permissible interference signal for wrong side failure

For wrong side failures the worst case fault in the traction return system, resulting in the “maximum” transfer function F_{max} , shall be assumed.

Let the interference signal which causes the track circuit to show clear while occupied be denoted by I_{TDSclear} .

Then the total permissible value of interference $I_{\text{RStot, clear}}$ for this case is given by:

$$I_{\text{RStot, clear}} = I_{\text{TDSclear}} / F_{\text{max}}$$

for all combinations of amplitude, frequency and duration of the interference signal.

5.6 Wheel detector susceptibility

Due to the principle of discrete detection of wheels passing a wheel detector, transient and continuous interference may be considered as equivalent.

The susceptibility of wheel detectors shall be determined under laboratory conditions as follows:

5.6.1 Set up a wheel detector arrangement as in the field with worst case geometrical conditions (e.g. worn wheels, small wheel diameter, worn rail profiles).

5.6.2 Inject a defined interference current into the rail with the wheel detector mounted, or apply a defined interfering electromagnetic field, representing the rolling stock (traction current and harmonics), and measure the resulting change of internal signal (current, voltage or frequency), which is responsible for the switching function of the wheel detector.

The frequency of harmonics shall include the operating frequency of the wheel detector.

The value of external interference signal causing an unwanted reaction of the wheel detector, both with and without a wheel passing, shall be measured.

For axle counters, it shall be shown that the reliability of counting will be maintained in spite of the expected interference. A differentiation between right side and wrong side counting failures due to interference is not required.

5.7 Train detection system gabarit

The train detection system gabarit is the maximum permissible value of interference signal determined above, reduced by the factor of safety (see A.10).

5.8 Interference signal generated by rolling stock and substations

The interference signal generated by rolling stock and substations is a function of the following:

- the interference signal generated by each source;
- the number of sources;
- the rules adopted for the summation of interference signals from different sources.

5.8.1 Interference signal generated by each source: track circuits

Track circuits are affected by the following sources of interference current:

- the current drawn by the rolling stock
 - with the rolling stock considered as passive impedances, and
 - with the traction power supply interference voltages superimposed on the supply voltage.

These currents are limited by setting a minimum impedance value for the rolling stock and by limiting the interference level generated by the substation. Under some circumstances the impedance of the rolling stock may interact with the impedance of the power supply to generate additional interference which is not attributable to either source alone;

- the current generated by the rolling stock powered with a “pure” voltage (d.c. or undistorted sine wave a.c.). This current includes:
 - the interference current generated by the traction equipment;
 - the interference current generated by on-board converters for auxiliary power supplies (note that the return current from these may flow through the rails beneath the train);
- unwanted coupling from the transmitter(s) of other track circuit(s).

5.8.2 Interference signal generated by each source: wheel detectors

The major sources have been defined in 5.3.4. The effect of multiple source interference shall be considered in the particular case.

5.9 Test report

The tests and the context in which they were performed shall be presented in a report, the main text and appendices of which should include the following.

5.9.1 Introduction

A general presentation of the systems under test.

5.9.2 Test organisation

A statement of who performed the tests and the contact address.

5.9.3 Configuration

A definition of the design status of the train detection system, including the status of hardware and software listed in the documentation of the factors affecting the train detection system's characteristics.

5.9.4 Reference documents

This includes the test plan, the description of the train detection system and the document listing the factors affecting its characteristics.

5.9.5 Application of test plan

With specific reference to compromises or amendments to the test plan which were found to be necessary, including

- test conditions - technical characteristics of the test site,
- instrumentation - a block diagram of the equipment used, the location of the measuring instruments, the interconnections between them, their accuracy, response characteristics, sensitivity, signal scalings, etc.,
- test procedure - calibration, verification of environmental noise, operational conditions of the system under investigation during tests.

5.9.6 Test results

These include

- the sensitivity of the train detection system alone,
- the transfer function of the train detection system as installed,
- the electrical equivalent circuit of the train detection system and traction power supply system,
- description of model (if used),
- verification of model (if used),
- results of tests; "normal" and "maximum" transfer functions,
- list of possible fault conditions considered,
- gabarit of train detection system,
- summation rules (for use by rolling stock operator).

5.9.7 Comments

An evaluation of the results, their validity (e.g. why the site was chosen) and comparison with expected results.

5.9.8 Archive of test results

Measurements often entail the collection of large quantities of recordings. It is not always practical for these to be reproduced and circulated with a report, but provision shall be made for their archive and reference made in the test report as to how authorised access to this documentation may be achieved.

6 Characterisation of rolling stock

6.1 Objectives of procedure

Tests shall be made on rolling stock to verify the interference generated and also for the following reasons:

- to acquire additional information concerned with the characterisation of the train power system;
- to compare measured levels of interference with those predicted to confirm the understanding of the train power system.

Although the tests will be made with reference to the specific routes on which the rolling stock will operate, tests should be defined, carried out and the results documented without ambiguity to avoid unnecessary repeat testing in the event that a future new route comes under consideration.

The acceptance of previously accepted rolling stock on new routes may not require further testing if the characteristics of the new route are similar to an existing route, e.g. an extension of an existing line with similar train detection and traction supply systems.

Electrical multiple unit trains shall be tested complete as a fixed formation; locomotives shall be tested separately from the coaches they haul. Each type of hauled passenger coach shall be tested separately in accordance with UIC 550. For summation rules for harmonic currents see A.11.

6.2 Description of rolling stock and factors affecting its characteristics

The factors on which the rolling stock / signalling compatibility depends shall be defined and documented by the manufacturer or operator and shall include all relevant modifications to the rolling stock. Safety critical components shall be clearly identified.

Annex C lists known factors which can affect the rolling stock / signalling compatibility.

6.3 Configuration (design status)

Measurements shall be carried out on a representative sample or samples of the rolling stock for which acceptance is to be obtained. The equipment shall be in a fully developed and modified operating state; the status of hardware and software which are factors affecting rolling stock / signalling compatibility, shall be known and fully documented.

6.4 Test plan

The test plan is a document which should be approved by the accepting body as an appropriate means of measuring the rolling stock characteristics. It shall make reference to the following criteria.

6.4.1 Test site

Measurements shall be carried out with the rolling stock operating on sections of track which are as representative as possible of those of the intended routes. The extent to which a non-totally representative test site may be used to explore the range of tests listed in 6.4.3 should be evaluated.

Rolling stock shall be tested on lines electrified with the different types of traction power supply with which the rolling stock can operate.

To avoid ambiguity of results, attention shall be made to the contribution of measured levels of interference resulting from the traction power supply characteristics and those actually generated by the rolling stock under test.

6.4.2 Instrumentation

Measurement and testing equipment shall be specified and agreed (in particular transducers, spectrum analysers, selective voltmeters).

The choice of instruments used and their set-up shall be such that measurements made can be interpreted with respect to the reference gabarit(s). Specifically, consideration should be given to

- location of measuring transducer,
- accuracy,
- bandwidth,
- dynamic range,
- quantification of systematic errors; e.g. noise, intermodulation effects, etc.,
- system response to evaluate signals within both frequency and time domains,
- measurement of modulation between frequencies or other gabarit specified criteria,
- recording of speed, location, traction power supply parameters and other factors which may have an influence on the level of interference currents generated,
- procedural calibration checks.

Annex B details ways in which the above have been addressed historically.

6.4.3 Test procedure

Tests shall be conducted over the range of conditions which may occur under normal operation with a sufficient number of tests being performed to enable the true rolling stock characteristics to become apparent. The worst cases shall be examined in the closest detail. Tests should be conducted in the following operational conditions:

- at standstill;
- the full extent of the effort-speed characteristics in motoring and braking (including regenerative braking);
- constant speeds up to the maximum (regulated by speed control or the driver);
- operation at reduced efforts as determined by the position of the driver's power controller;
- sequences of motoring, coasting and braking;
- normal and degraded modes of operation;
- typical variations or disturbances in the supply voltage (e.g. due to other trainsets on the system, a substation cut out, line gaps, poor contact of current collection equipment etc.);
- environmental conditions which can affect the rolling stock equipment operation (e.g. wheelslip, wheelslide etc.);
- regular known transients (e.g. circuit breaker opening / closing, particular rolling stock equipment starting / stopping).

6.5 Test report

The tests and the context in which they were performed shall be presented in a report, the main text and appendices of which should include the following.

6.5.1 Introduction

A general presentation of the systems under test (rolling stock, power supply system and train detection system).

6.5.2 Test organisation

A statement of who performed the tests and the contact address.

6.5.3 Configuration

A definition of the design status of the rolling stock, including the status of hardware and software listed in the documentation of the factors affecting the rolling stock's characteristics.

6.5.4 Reference documents

This includes the test plan, the description of the rolling stock and the document listing the factors affecting rolling stock's characteristics.

6.5.5 Application of test plan

With specific reference to compromises or amendments to the test plan which were found to be necessary, including

- test conditions - technical characteristics of the test site,
- instrumentation - a block diagram of the equipment used, the location of the measuring transducer, the interconnections between instruments, their accuracy, response characteristics, sensitivity, signal scalings, etc.,
- test procedure - calibration, verification of environmental noise, number of test runs, operational conditions of the system under investigation during tests.

6.5.6 Test results

An analysis and summary of the measurements made with typical examples of recordings.

6.5.7 Comments

An evaluation of the results, their validity and comparison with expected results.

6.6 Archive of test results

Measurements often require the collection of large quantities of recordings. It is not always practical for these to be reproduced and circulated with a report, but provision shall be made for their archive and reference made in the test report as to how authorised access to this documentation may be achieved.

7 Characterisation of traction power supply systems

7.1 Objective

The objective is to determine the influence of the power supply system upon the characterisation of the rolling stock.

Relevant factors include but are not limited to the following:

- voltage tolerance;
- frequency tolerance;
- harmonic content;
- transients;
- maximum MVA rating;
- impedance of substation;
- impedance of catenary or conductor rail(s);
- impedance of return system;
- normal and degraded modes of operation.

Information about power supply characteristics are defined in EN 50163.

Since certain interference frequencies may be generated by the substation or by the rolling stock, or both, there may be ambiguity as to the source of these frequencies. This ambiguity shall be resolved. Resonances and oscillations shall also be considered.

7.2 D.C. traction power supplies

D.C. traction power supplies are, due to their rectifiers, particularly likely to produce interference currents which can affect track circuits. Annex D briefly describes the interaction of interference currents between the rolling stock and the d.c. traction power supply.

7.2.1 Test procedure

7.2.1.1 Substations equipped with diodes only

The voltage ripple is mainly due to the rectifier bridge and to phase unbalance in the high voltage supply. As a first approach, the voltage ripple should be measured on board the rolling stock in the neighbourhood of the substation with the traction power system cut out. Additionally, in the case of major ambiguity, the voltage ripple shall be measured on a power resistive load after agreement between the railway infrastructure authority and the organisation conducting the tests.

7.2.1.2 Substation equipped with regulated converters

Additionally, the voltage ripple may be due to unbalances in the control system of the converters. The voltage ripple shall be characterised at no load and on a power resistive load.

7.3 A.C. traction power supplies

Refer to EN 50163.

Annex A (informative)

Guidelines for the determination of susceptibility of train detection systems

A.1 Examples of system configurations

Some examples of train detection system configurations are shown below. They are simplified and do not show, for example, structures bonded to the traction return rail. It is emphasised that these are not the only possible configurations; they are included as examples of the type of situation that should be considered.

A.2 “Normal” configuration

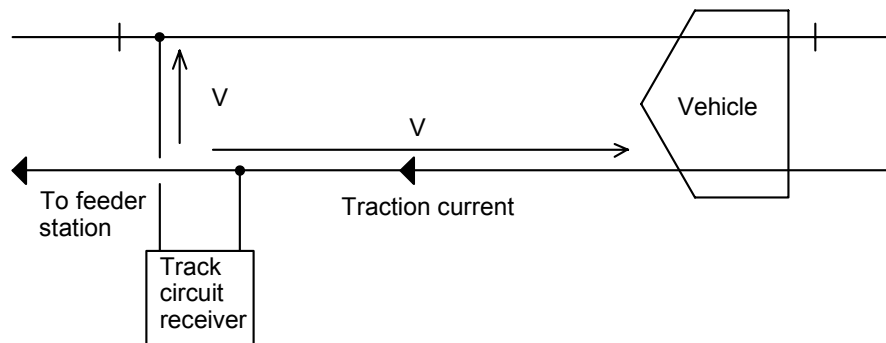


Figure A.12 – Interference mechanism with rails intact

The voltage drop along the length of traction return rail included within the track circuit appears across the track circuit receiver when the track circuit is occupied.

A.3 Interference mechanism with broken signal rail

With a break in the signal rail the track circuit will show occupied (a “self-revealing” fault). The circuit is as shown:

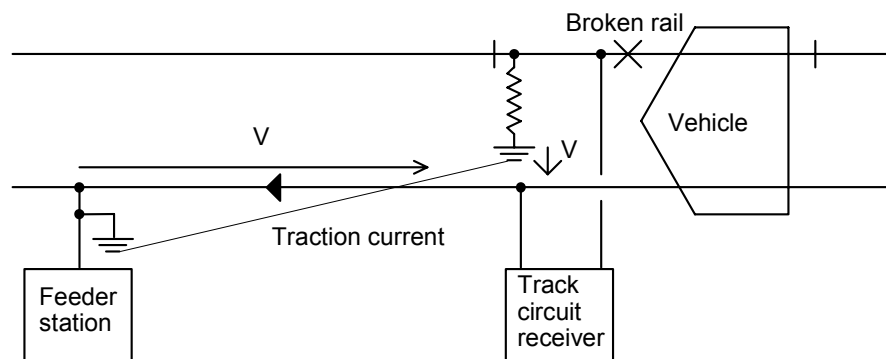


Figure A.13 – Interference mechanism with self-revealing broken rail

The voltage appearing at the receiver is now determined by the voltage drop between the vehicle and the feeder station, although the ballast resistance of the broken rail is in series with the input of the track circuit receiver. Since the broken rail causes the track circuit to show occupied, this situation should arise very infrequently.

A.4 Interference mechanism with broken return rail

In the case of an unrevealed break in a traction return rail, the circuit becomes:

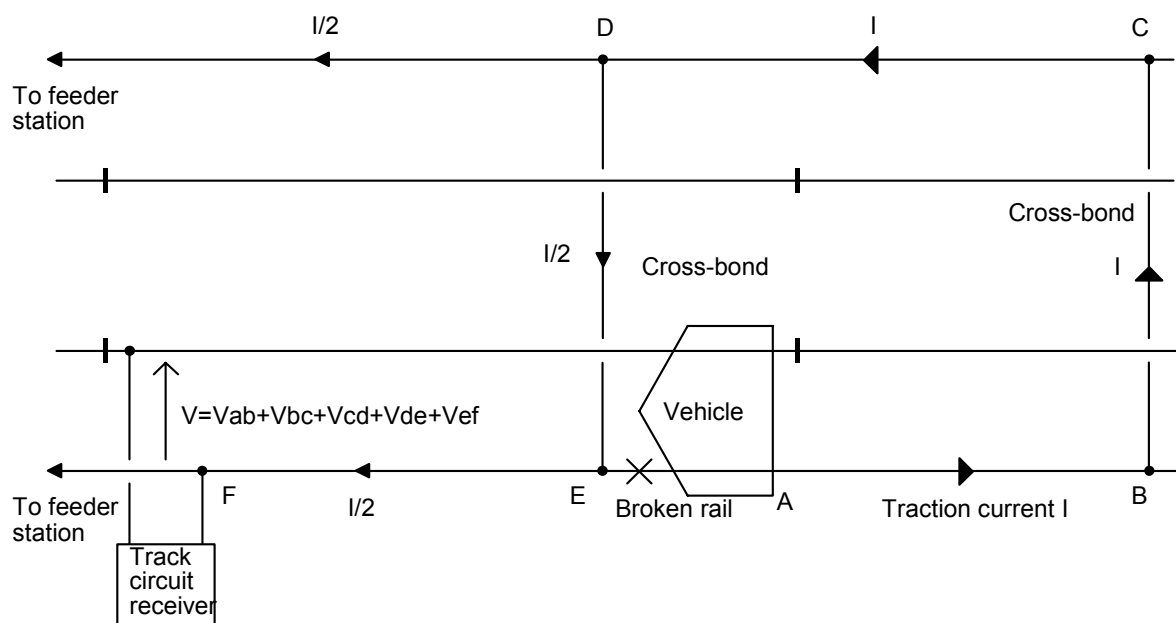


Figure A.14 – Interference mechanism with unrevealed broken rail

Since the traction current now has to take a longer return path to the feeder station, (although a proportion will flow in the earth), the interference voltage at the receiver will be increased. The length of the return path depends on the spacing of the cross-bonds and the length of the track circuit.

A.5 Double rail track circuits

A typical configuration for a double rail track circuit is shown:

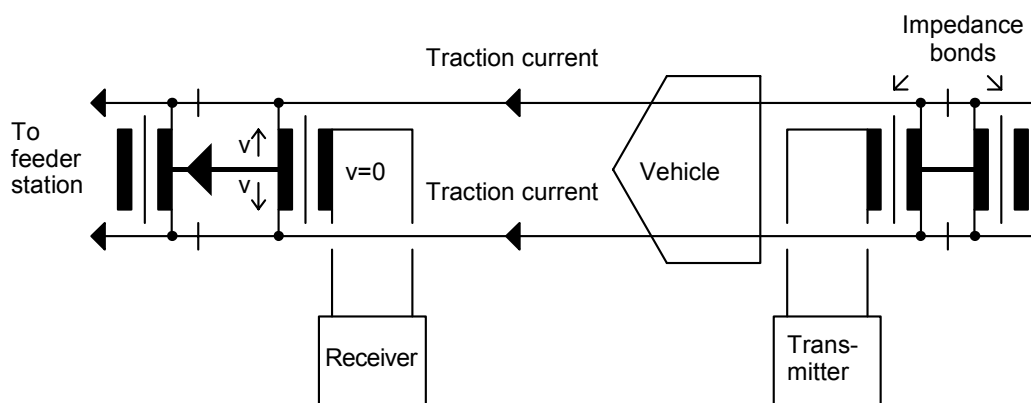


Figure A.15 – Double rail track circuit

In normal operation the traction return current in each rail is approximately the same and little voltage is developed at the track circuit equipment terminals. However in the case of a broken rail or impedance bond, the return current will be totally unbalanced resulting in a transverse voltage:

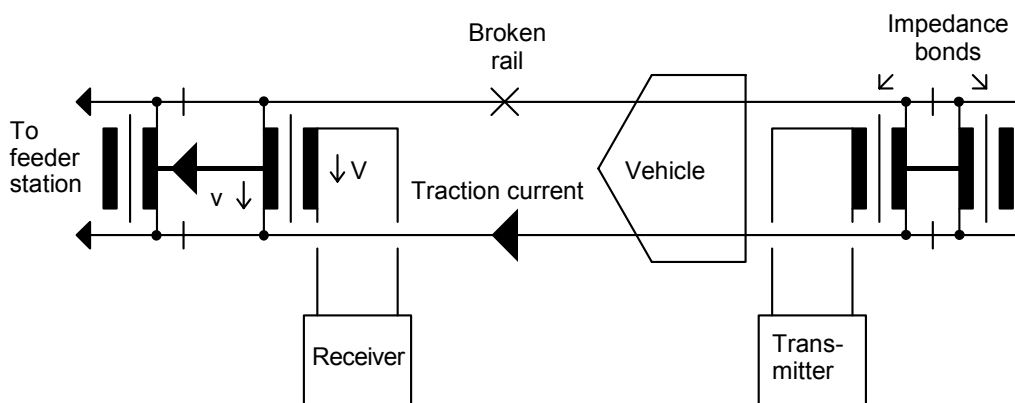


Figure A.16 – Double rail track circuit with broken rail

Such a break will normally cause the track circuit to show occupied (self-revealing) but depending on the type of track circuit it could be possible for the receiver to be falsely energised by interference before the train enters the track circuit, thereby permitting the train to proceed. The track circuit could then continue to show clear when it is occupied by the train.

A.6 Voltage between axles of rolling stock

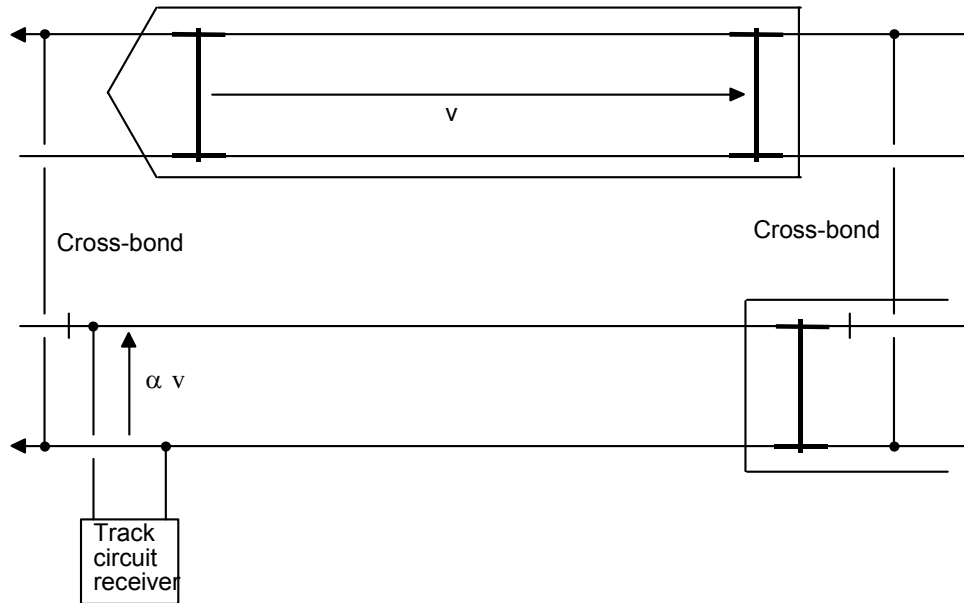


Figure A.17 – Interference mechanism due to voltage between axles – Case 1

If a voltage is generated between the axles of the train, a proportion α of this voltage may appear across the track circuit receiver. If the train length is similar to the track circuit length and to the cross-bond spacing, α may approach unity.

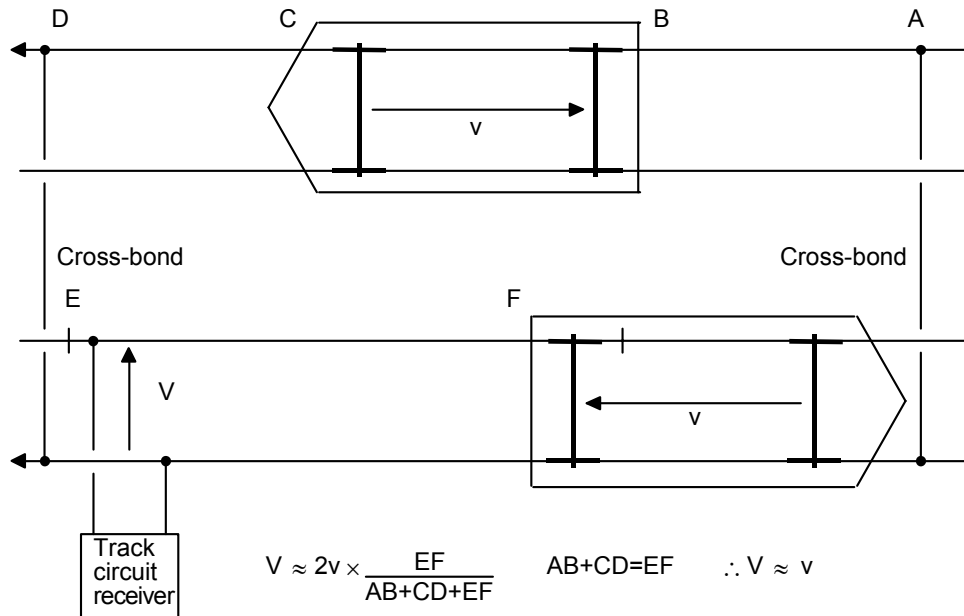


Figure A.18 – Interference mechanism due to voltage between axles – Case 2

If two trains contribute to the interference, the combined effect is similar to that for one train.

A.7 Effect of resistance between coupled vehicles

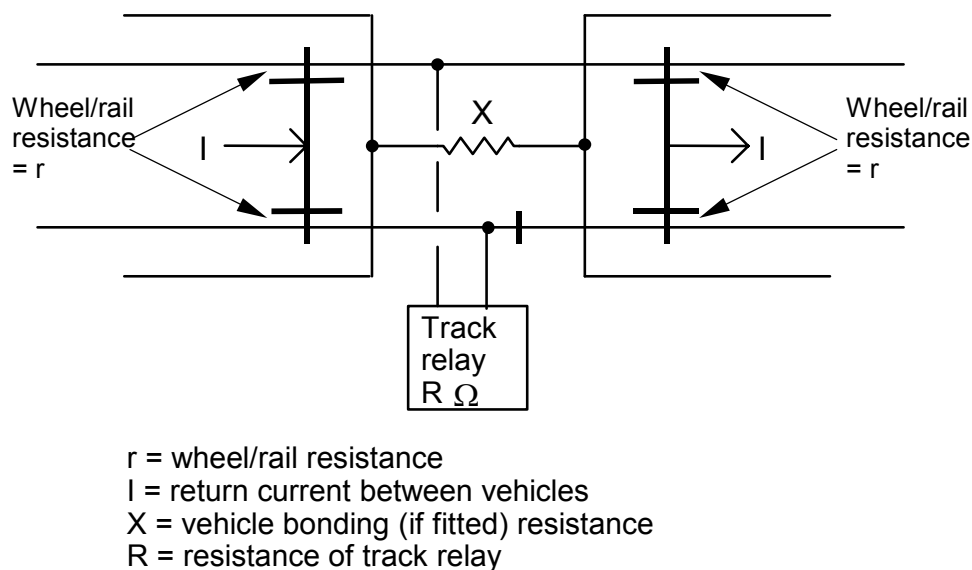


Figure A.19 – Effect of inter-vehicle current

This can be re-drawn (ignoring rail resistance):

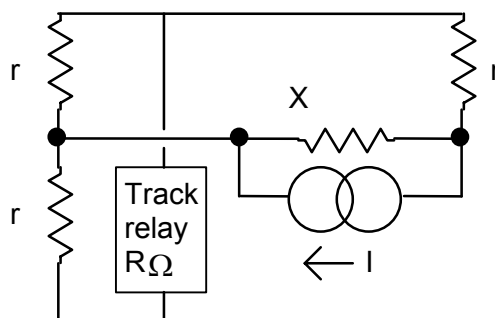


Figure A.20 – Equivalent circuit for previous figure

If all r are equal
 and $R \gg r$
 and $X \rightarrow \infty$ (either disconnected or not fitted)
 then $V_R \rightarrow I \times r$

If $X \approx r$
 then $V_R \rightarrow (I \times r)/3$

If the maximum tolerable V_R is 200 mV and I is 120 A then

$r_{\max} = 1,7 \text{ m}\Omega$ if $X \rightarrow \infty$
 or
 $r_{\max} = 5 \text{ m}\Omega$ if $X \approx r$

These values are quite low.

A.8 Radiated interference

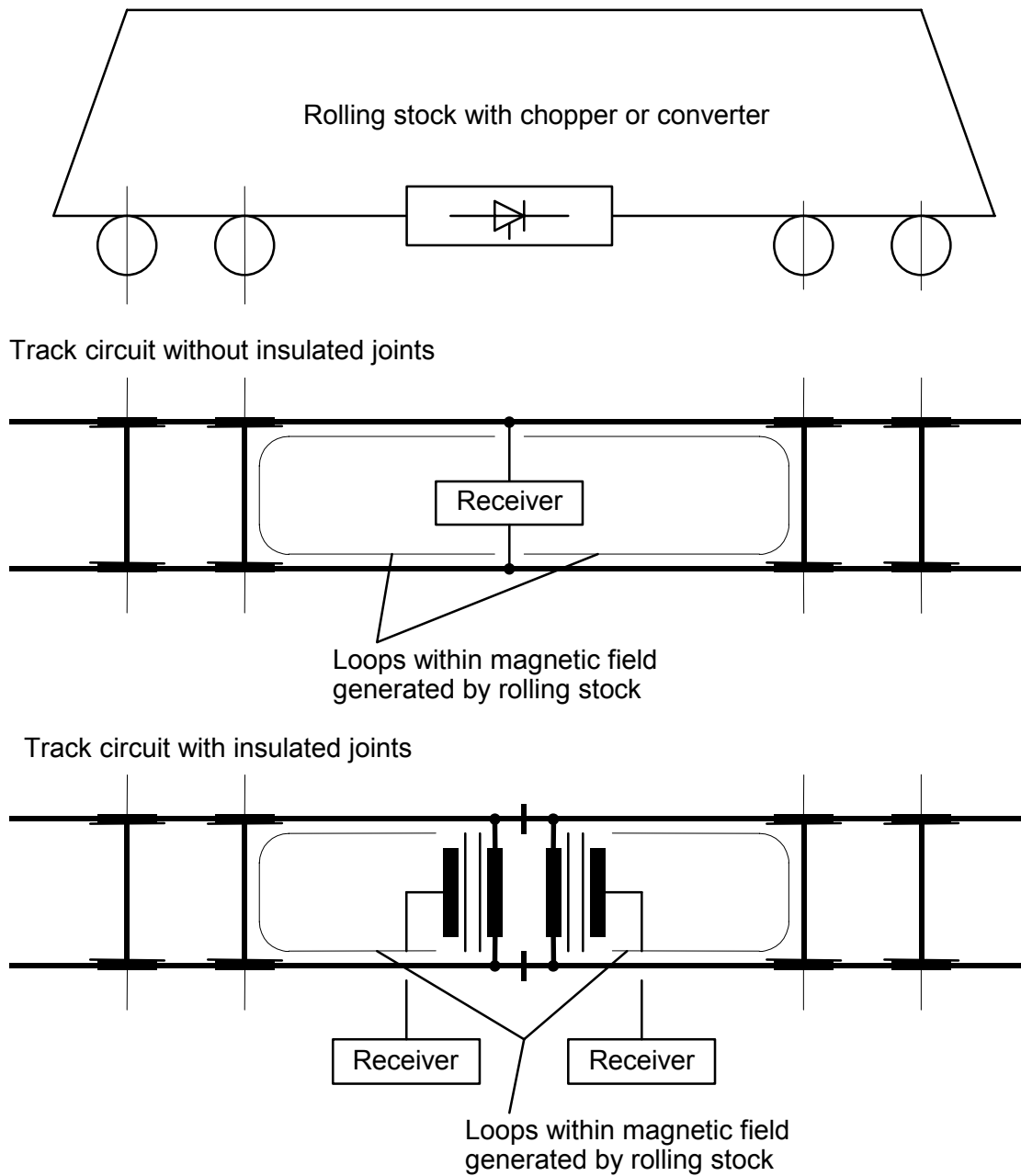


Figure A.21 – Example of radiated interference

A.9 Example of sensitive zone of wheel detector

Figure A.11 below shows the area in which mechanical parts of vehicles may have an influence on wheel detectors.

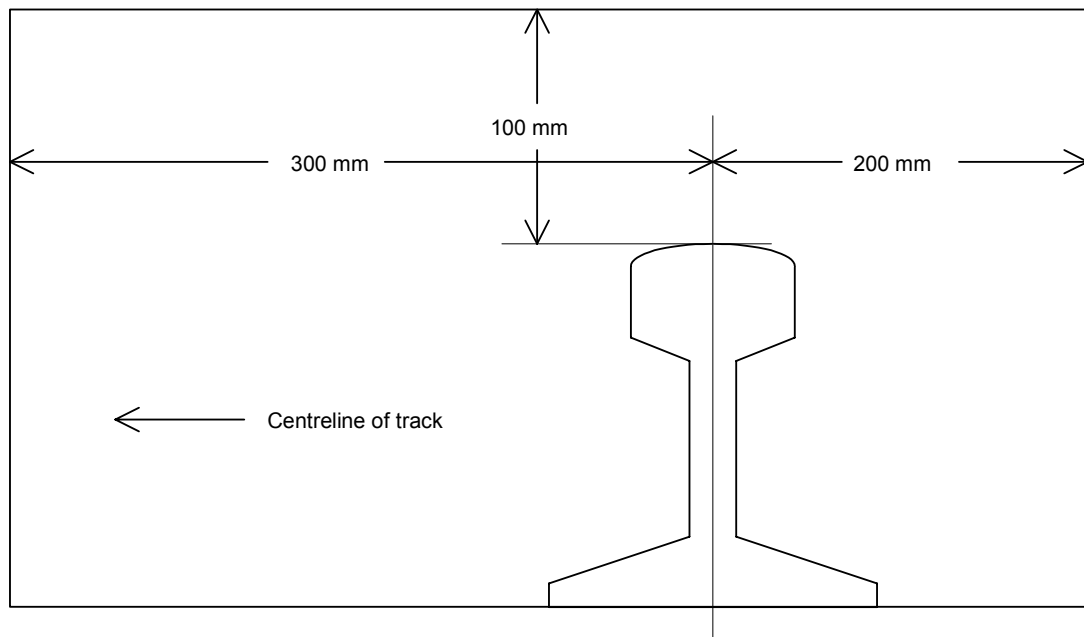


Figure A.22 – Sensitive zone of wheel detector

A.10 Factor of safety

The consequences of an unsafe failure of a train detection system depend on the way the indication given by the system (track occupied or clear) is used in the interlocking. For example the effects of a false clearance could be greatly reduced by the use of sequential proving of track section occupation in the interlocking.

The value of the factor of safety will be chosen according to the following considerations:

- accuracy of known information (e.g. track layouts, number of trains);
- possible background interference levels;
- accuracy of measurements;
- accuracy of predictions (e.g. computer models);
- “safe” or “unsafe” failure of the train detection system;
- the way the train detection system indication is processed in the interlocking;
- established national practices.

For these reasons it is not possible to specify in advance what factor of safety should be applied. It should be determined and justified as part of the compatibility case for each proposed application (see Clause 4). In countries where a fixed factor of safety is normally applied, this should be adopted, unless the compatibility case shows that a higher value is required.

Recommendations for factors of safety are given in UIC 737-3.

A.11 Multiple interference sources

The maximum number of interference sources that could affect a given train detection system at any one time should be determined from operational considerations. Summation rules should take into consideration the interdependence between multiple equipments.

ORE B108/1 established rules for the distribution of interference current from the traction equipment and from auxiliary converters etc. depending on the position of the train in relation to the track circuit. These general rules are superseded by the specific transfer function established under this procedure.

Annex B (informative)

Guidelines for the measurement of rolling stock characteristics

B.1 An example of a conceptual block diagram of a system used to measure interference currents

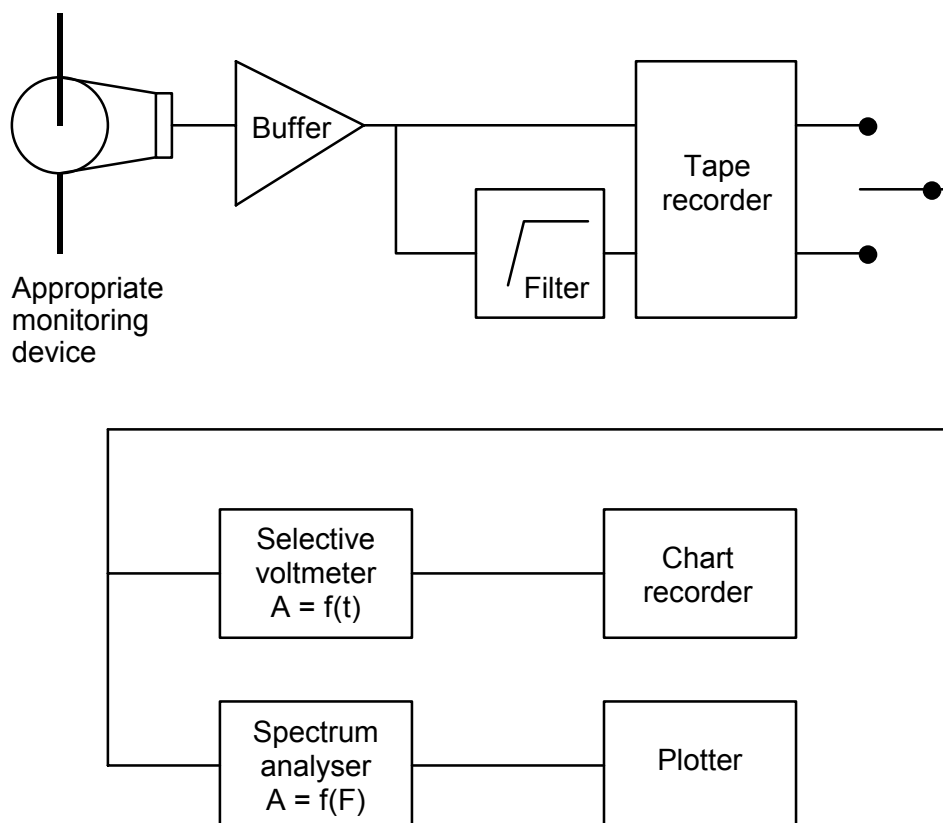


Figure B.2 – Example of system for measurement of interference currents

The choice of the interference current measuring depends on the type of rolling stock:

- for electrical locomotives and electrical multiple units the measurements are taken as close as possible to the current collection equipment; pantograph or 3rd rail shoe gear. Exceptionally the measurement may be made elsewhere in the current path provided it can be demonstrated either theoretically or practically, that there will be no significant difference in the results;
- for diesel locomotives, the measurements are made on the high voltage auxiliary train line;
- for trailer equipment (for example a static inverter or battery charger) measurements are made either on the High Voltage auxiliary train line, in the case of various elements connected in parallel or at the equipment input if a single element is involved.

Transducers to be used for measurement are suitable for the type of supply, (single phase AC or DC) and are specified with sufficient precision to allow the range of frequencies of interest to be adequately explored. The characteristic response of the measuring transducer is known from a calibration test record.

The equipment used in the instrument system processes the signal from the transducer to simulate the characteristic response of the susceptible track circuits. In particular:

- all relevant signals are recorded in a mass memory medium such as magnetic tape;
- the gain and pre-filtering of the system are adjusted to obtain the best dynamic range for the frequencies and levels to be investigated, taking into account the limitations on dynamic range of recorded signals;
- the interference current is signal processed by one or several instruments adapted to simulate the track circuit characteristics and output on to a graphic recorder or plotter. The signal processing may involve analogue, digital or a combination of both techniques and specific analysis, such as modulation between frequencies, which may be specified in the gabarit under investigation;
- the interference current is signal processed as function of frequency and recorded with one or more spectrum analysers;
- preferably the plotting of processed signals is done in real time as tests proceed, but if the signal processing required is too complex, post analysis of recordings may be the only alternative;
- a measurement of the overall background noise is made by routing the current supply outside the measurement transducer;
- the overall instrumentation system is tested by injecting signals of relevant frequencies and amplitude either directly into the measuring transducer or as close to the input of the instrumentation system as possible;
- environmental signals and signals indicating the operation condition of the trainset are recorded with the interference current signal. These normally include the supply parameters, the speed, the location, whether the trainset is motoring, braking or coasting, etc.

Annex C (informative)

Factors affecting rolling stock characteristics

C.1 Factors affecting compatibility

The factors which are known historically to be relevant are as follows:

- individual vehicle and overall train lengths;
- spacing between the individual wheelsets of each vehicle and between the outermost wheelset and the vehicle extremity;
- the construction of wheelsets and the method of train braking;
- the electrical bonding between the running surface of wheels and the vehicle bodyshell, also the electrical bonding between vehicles;
- the traction and auxiliary power circuits (including the train line return current); the filter values, return current paths, interaction between power circuit elements;
- the traction and auxiliary control packages; chopping frequencies, hardware and software control techniques, feedback transducers, fault detection;
- current collection equipment, distance between line contact points;
- interference current monitoring units and other equipment that may be used to monitor signalling interference related parameters;
- magnetic fields produced by vehicles;
- minimum maintenance requirements;
- different operating conditions and degraded modes of operation where the other factors listed may change.

Annex D (informative)

D.C. traction power supplies

D.1 General

A d.c. substation changes the a.c. supply from a public distribution system to d.c. by means of a transformer and a converter (usually a rectifier), sometimes with the addition of a filter.

Figure D.1 shows the rolling stock and substation system.

The traction current flowing in the rails contains interference currents generated by the rolling stock, and by the power supply system as described below.

D.2 Interference currents generated by the rolling stock

Figure D.2 gives an equivalent circuit diagram for interference currents flowing from the rolling stock, considered as a current source, to the substation, considered as a receiver, through the catenary and the return circuit. The maximum value of the interference current depends on

- the current ripple generated by the rolling stock,
- the input filter characteristic of the rolling stock,
- the impedance of the substation filter (= 0 if there is none),
- the impedance of other rolling stock,
- the impedance of the catenary or third rail,
- the impedance of the traction return system.

D.3 Interference currents generated by the traction power supply system

Figure D.3 gives an equivalent circuit diagram for the interference currents circulating from the substation, considered as a generator, to the rolling stock, considered as a receiver.

The maximum value of the interference current depends on

- the characteristics of the substation filter,
- the voltage ripple generated by the substation converter - the whole of this voltage ripple is present at the substation output if there is no filter,
- the input impedance of the rolling stock,
- the impedance of the catenary or third rail,
- the impedance of the traction return system.

The voltage ripple generated by the substation converter is mainly due to

- the public distribution system (voltage imbalances, loss of voltage, harmonics caused by other customers, etc.). The limits for these disturbances are given in EN 50160,
- the transformer,
- the converter.

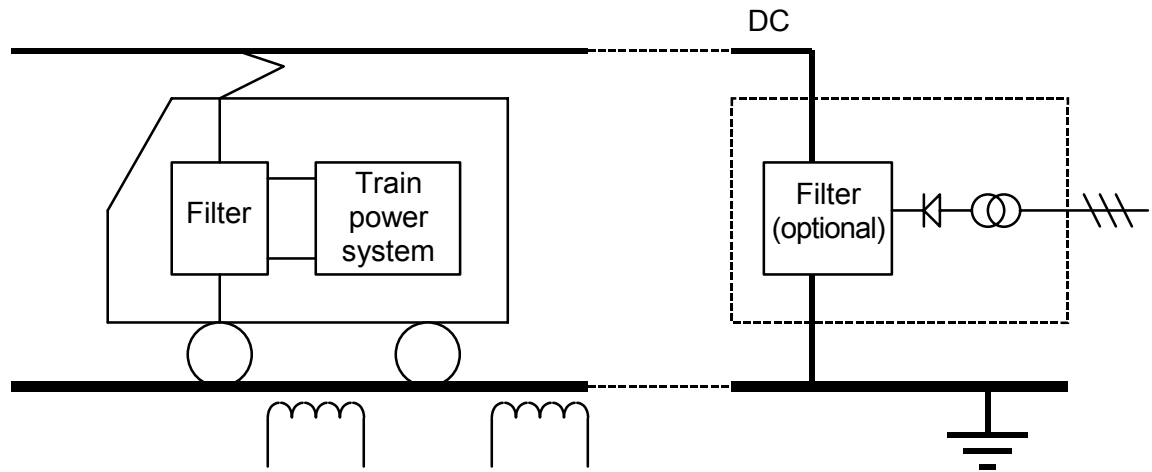


Figure D.4 – Rolling stock with DC supply

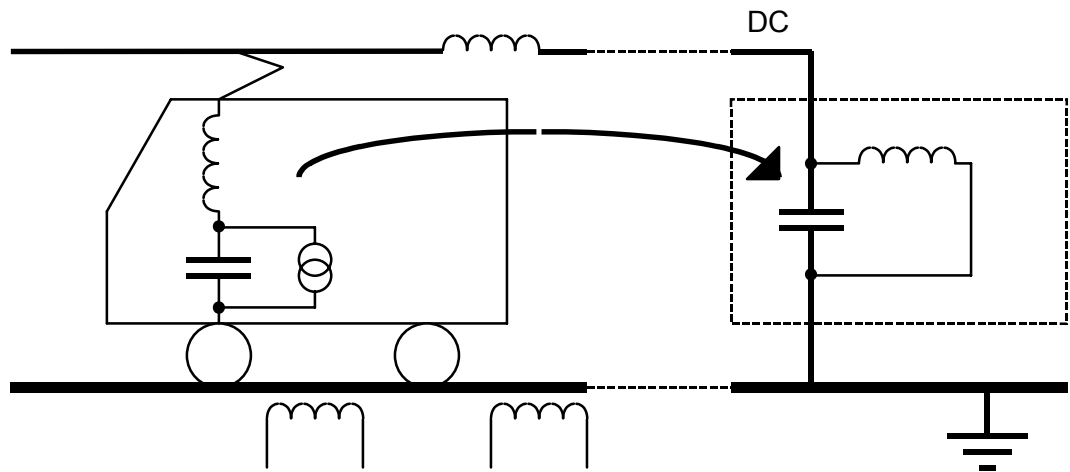


Figure D.5 – Circulation of interference current generated by rolling stock

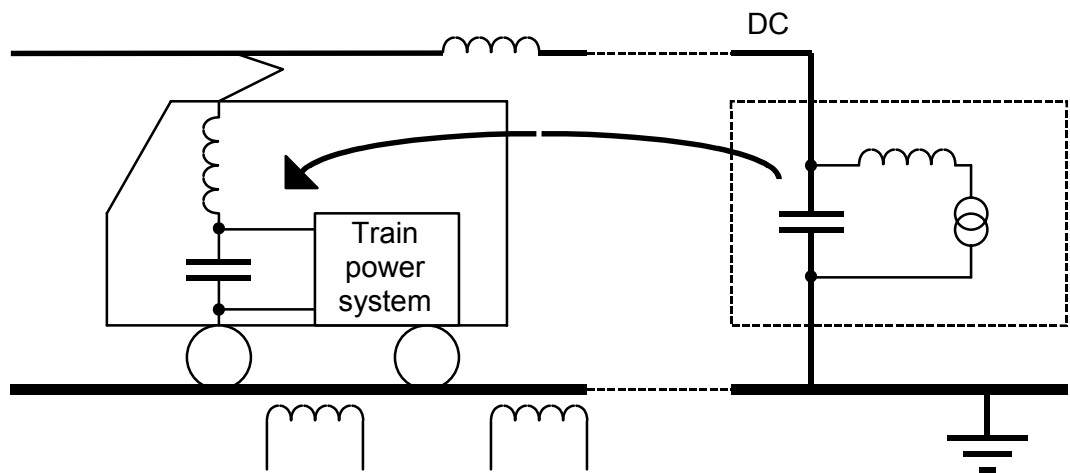


Figure D.6 – Circulation of interference current generated by the substation

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